Qubitsense

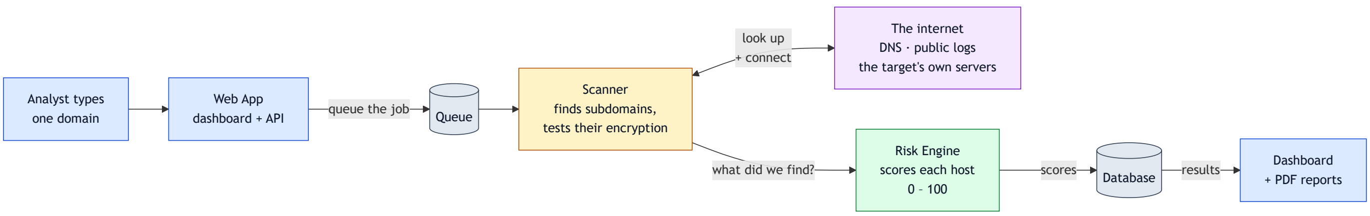
Finding the encryption that quantum computers will break

The problem, in one paragraph

Almost every website today protects its traffic with encryption that a large quantum computer would break. Attackers know this, so they **record encrypted traffic now and decrypt it later**, once the hardware exists. Security teams call this *Harvest Now, Decrypt Later*. A bank with hundreds of subdomains has no easy way to find out which of them are exposed, or which ones are worth harvesting.

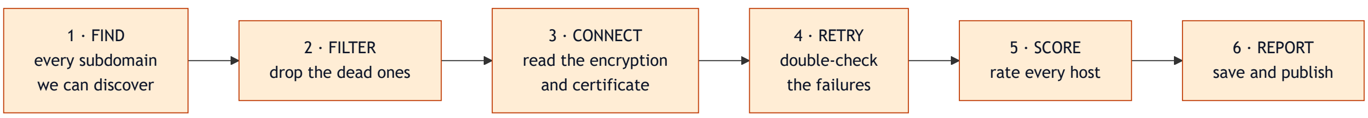
Qubitsense answers that. Give it one domain. It finds every subdomain the organisation owns, connects to each one, inspects the encryption actually in use, and returns a risk score for every host plus a single rating for the whole estate.

1. How the system works



The scanner runs as its own process, separate from the web app. That is why a scan of 200 subdomains never freezes the dashboard — the two only ever talk through the queue.

2. What one scan actually does



Step	What happens	Why it matters
1 · Find	Three sources at once: public certificate logs, a threat-intelligence feed, and a 200-word DNS guess list	One source alone misses most of an estate
2 · Filter	Throw away hostnames with no live DNS record	Stops us wasting a slow connection on a ghost
3 · Connect	50 connections at a time, 12-second limit each	200 subdomains finish in minutes, not hours

Step	What happens	Why it matters
4 · Retry	Anything that failed on the network is tried once more	A timeout is not the same as "no encryption"
5 · Score	Every host gets a 0–100 risk score	See section 3
6 · Report	Dashboard, PDF, and a machine-readable inventory	Findings you can hand to an auditor

> **If a scan finds nothing, it says so.** A domain that does not exist is marked **failed**, not quietly reported as a page of zeroes. This matters more than it sounds — an empty report and a catastrophic one look identical otherwise.

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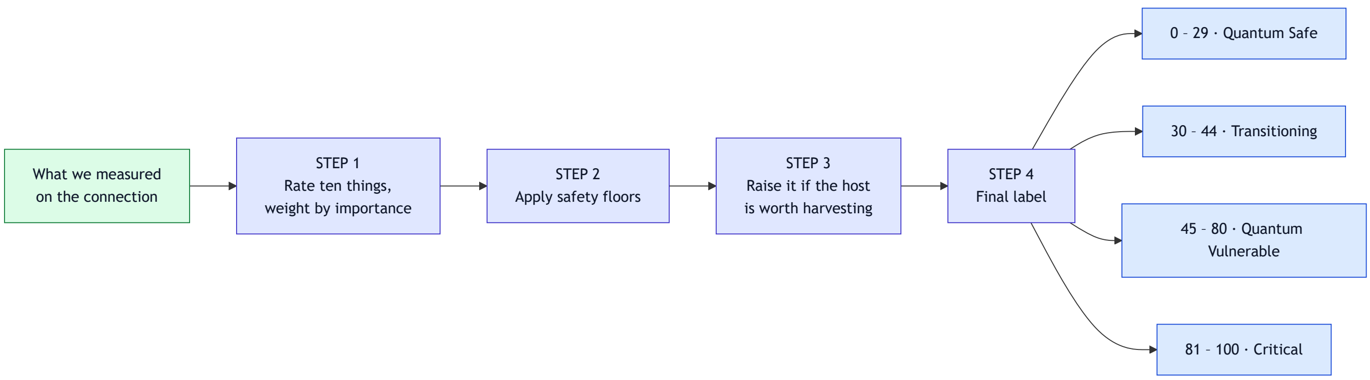
3. How a host gets its score

Every number comes from what was actually measured on the connection. Nothing is random, nothing is hardcoded, and the full rulebook is published by the app itself so anyone can check a score against it.



Step 1 — the ten things we rate. The weights add up to exactly 100%.

Weight	What we check	Plain meaning
22%	Key exchange	Can a quantum computer unlock the session key?
14%	Certificate signature	Could the certificate be forged?
12%	Key strength	How much work is breaking the key?
10%	Forward secrecy	If one key leaks, does *every* past session leak too?
10%	Protocol version	Is the protocol itself current?
8%	Cipher strength	Does the encryption survive a quantum search attack?
7%	Trust chain	Is the certificate expired, self-signed, or for the wrong name?
6%	Cipher mode	Is it a mode with known attacks?

Weight	What we check	Plain meaning
6%	Hash function	Is the fingerprint algorithm still safe?
5%	Certificate lifetime	Does this team rotate certificates properly?

Step 2 — the safety floors.

Situation	Score cannot go below
Already breakable today, no quantum computer needed	**81**
Nothing quantum-safe anywhere on this host	**45**
Half-way through migrating	**30**

Step 4 — the estate rating. Every host's score rolls up into one number from 0 to 1000 and a tier: *Elite-PQC*, *Advanced*, *Standard*, *Developing*, or *Legacy*.

Why the floors exist

This is the part most scoring tools get wrong. Imagine a server running the newest protocol, the strongest cipher, a modern hash and a valid certificate. It scores well on **eight of the ten things we measure** — but the two that decide whether a quantum computer can break it are still completely classical.

A simple average would call that server *healthy*. It is not. The floors force the honest answer: **if nothing on this host is quantum-safe, it cannot be rated better than "Quantum Vulnerable", no matter how tidy the rest of it is.**

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4. What you get at the end

Output	Who it is for
Live dashboard	Risk scores, the riskiest hosts, a map of the estate
PDF report	The whole scan, ready to attach to a review
Per-host fix list	One page per server: what is wrong, in priority order
Machine-readable inventory	An industry-standard list of every algorithm in use
Plain-English assistant	Ask questions about a scan and get answers from its real numbers

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5. What it is built with

Part	Technology
Web app and API	Python, FastAPI

Part	Technology
Scanner	Python `asyncio` — 50 connections at once
Storage	SQLite
Dashboard	HTML, Tailwind CSS, Chart.js
Assistant	Llama 3.3 70B via Groq
Login	Password + authenticator-app 2FA

Three programs run side by side: the web app, the scanner, and a scheduler for recurring scans. One command starts all three.

Deeper technical detail — every module, endpoint, and scoring constant — is in **ARCHITECTURE.md**.